

Improving Group SKR in HAP-Based Multi-User QKD Using Adaptive TDMA Under Dynamic Clouds

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1.Introduction

Background: Need for Secure Key Distribution

- More devices are connected across wide and remote areas.
- Remote areas such as mountains, islands, and offshore regions are difficult to connect with physical infrastructure.
- Future quantum computers may threaten conventional public-key cryptography.

New secure key-distribution technologies are needed.

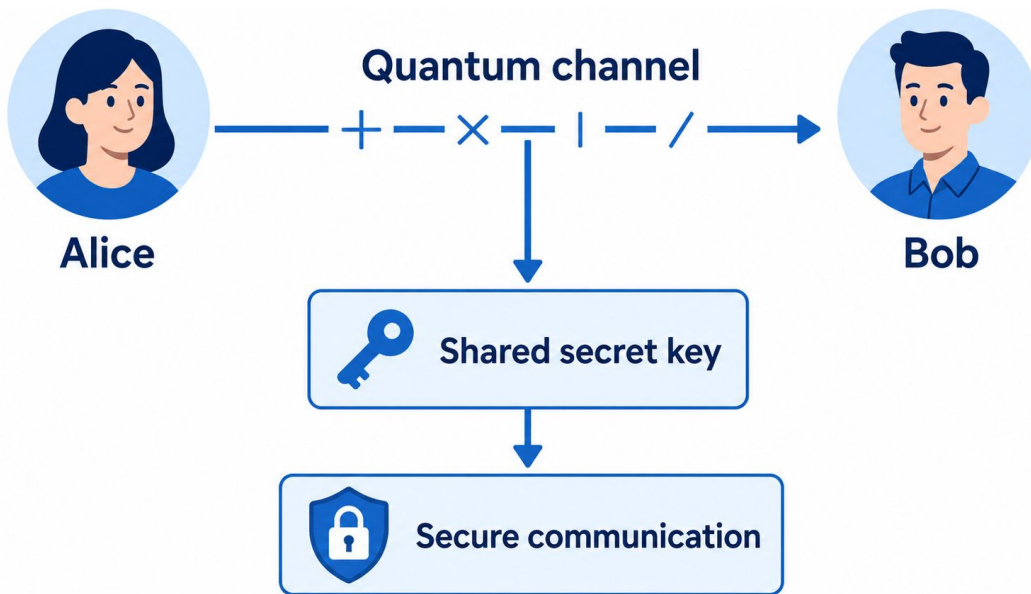
→ QKD is attracting attention.



What is Quantum Key Distribution (QKD)?

- QKD generates a shared secret key between legitimate users.
- It does not directly encrypt data; the generated key is used for symmetric encryption.
- Eavesdropping disturbs quantum states and can be detected through errors.

QKD is a method for securely generating shared keys, not a method that directly sends encrypted data.

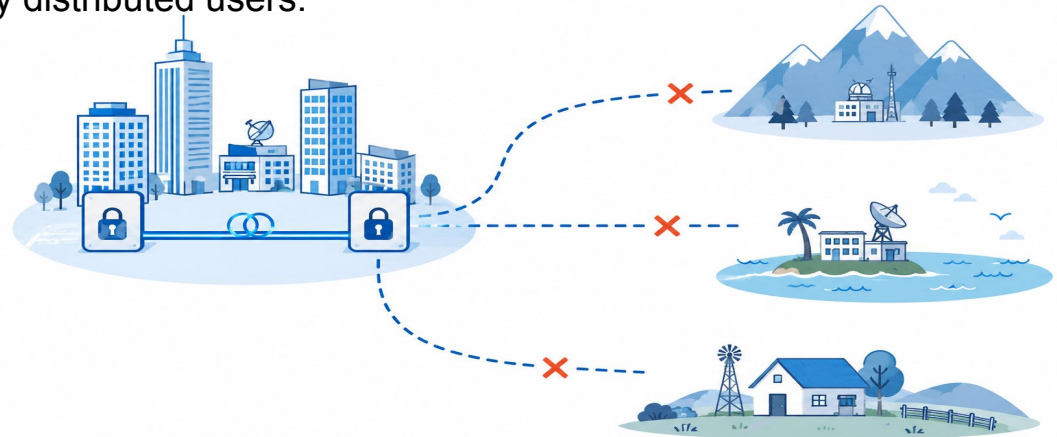


Research Motivation: Limitation of Fiber-Based QKD

Current QKD systems mainly rely on optical fiber links.

- Suitable for fixed networks between central locations
- Difficult to deploy in mountains, islands, and offshore areas
- Installing fiber to every remote user is costly and impractical

→ Fiber-based QKD is not suitable for widely distributed users.

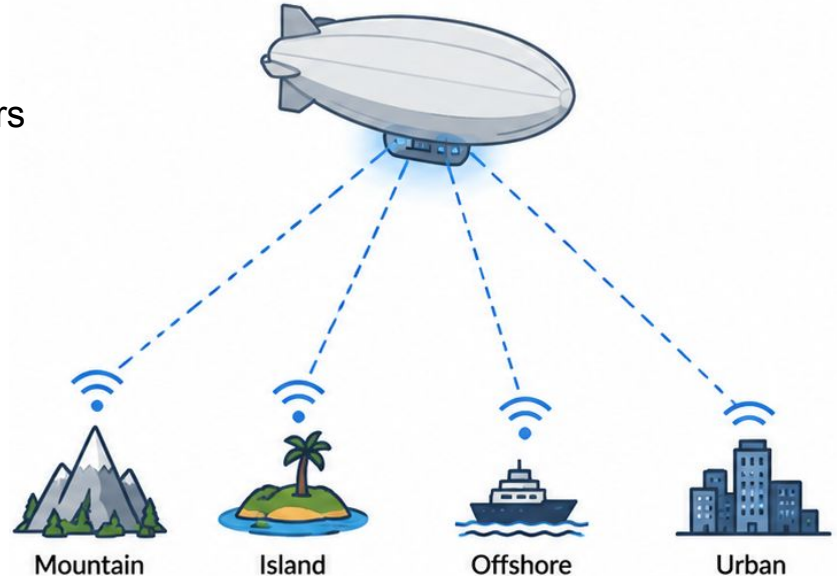


Proposed Solution: HAP-Based FSO QKD

Free-Space Optics (FSO) can replace physical fiber links.

- A HAP can cover wide and remote areas
- Ground users can be served without fiber deployment
- One HAP can distribute quantum keys to multiple users

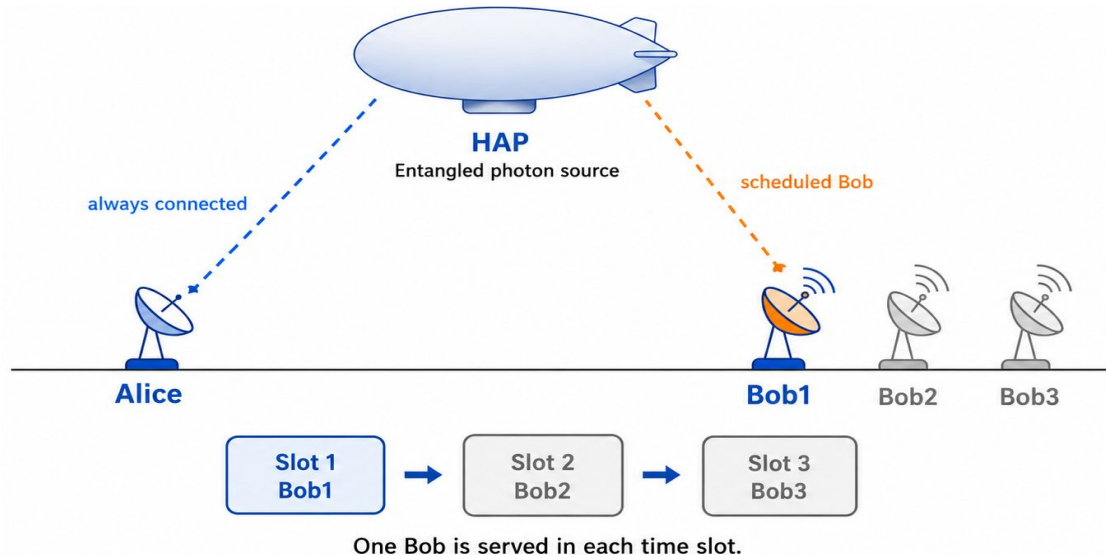
→ HAP-based FSO is promising for multi-user QKD.



HAP-Based Multi-User QKD Using BBM92 and TDMA

- The HAP generates entangled photon pairs.
- One photon is sent to Alice, and the other is sent to a scheduled Bob.
- In BBM92, matched measurement bases are used to generate sifted key bits.
- TDMA allows multiple Bobs to share the HAP–Bob transmission time.

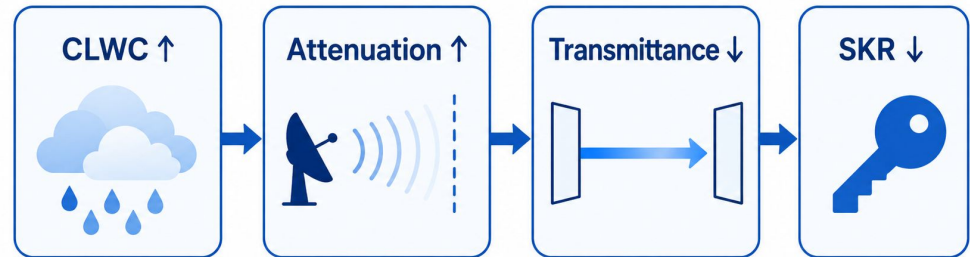
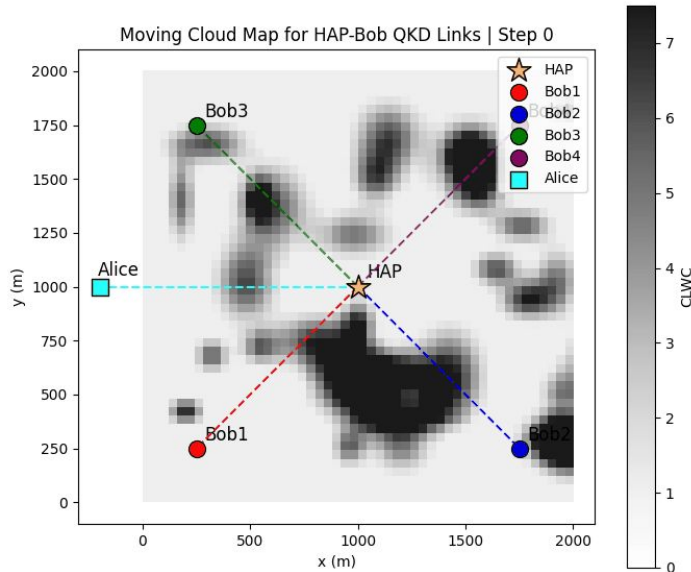
BBM92 generates keys from entangled photons, and TDMA enables one HAP to serve multiple Bobs.



BBM92: entangled photons + matched bases → shared key

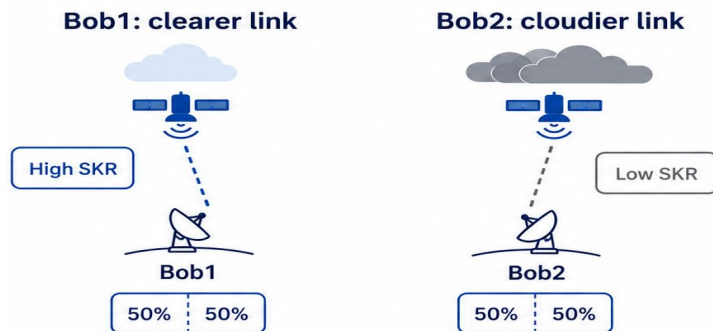
Dynamic Cloud Challenge in HAP–Bob FSO Links

- HAP–Bob FSO links are affected by clouds.
- Higher CLWC causes stronger attenuation.
- Cloud conditions differ for each Bob and change over time.
- Therefore, the SKR of each Bob is time-varying.



Motivation: Limitation of Fixed TDMA Under Dynamic Clouds

- Conventional TDMA assigns fixed time slots to users.
- Fixed TDMA does not consider time-varying cloud conditions.
- Even if Bob1 and Bob2 have different SKRs,
both users receive the same time allocation.
- The worse Bob limits the Group SKR.



$$t_1 = t_2 = 0.5$$

$$\text{Group SKR} = \min(0.5R_1, 0.5R_2)$$

Fixed TDMA: 50% : 50%



Research Questions and Goal

Main Goal:

Improve Group SKR under dynamic cloud conditions.

Approach:

- Estimate the cloud condition of each HAP–Bob link
- Allocate TDMA time according to channel quality
- Give more time to the worse link
- Avoid extreme allocation using CLWC capping and scaling

Research questions

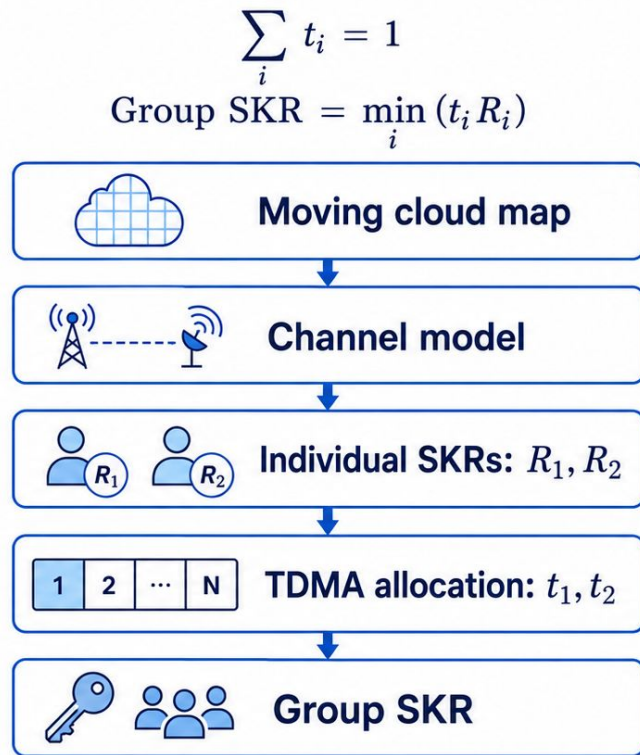
1. Can Adaptive TDMA improve Group SKR?
2. How does group size affect Group SKR?
3. Can temporary worst-Bob exclusion improve active-group SKR?

2. System Model and Adaptive TDMA Method

System Model and Evaluation Flow

- Initial evaluation: Alice, Bob1, and Bob2.
- HAP–Bob channel: cloud attenuation, turbulence, beam spreading, and misalignment.
- HAP–Alice cloud loss: 0 dB.
- (R_i): individual SKR, (t_i): TDMA time fraction.

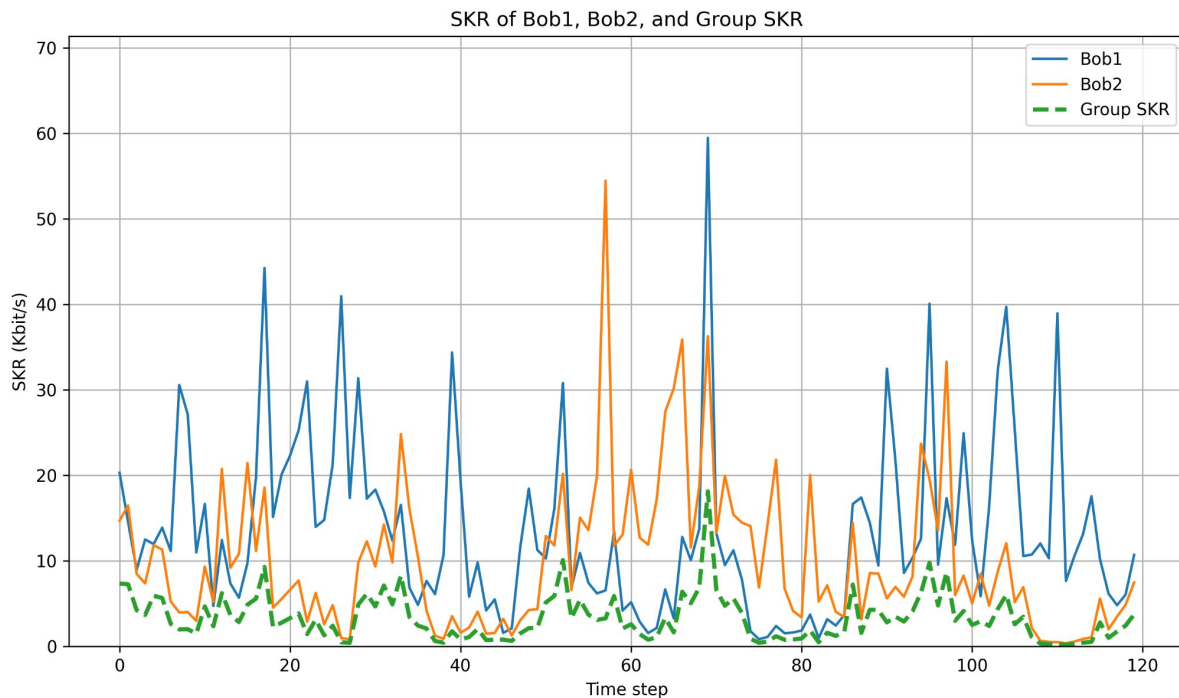
The cloud condition changes individual SKRs, and TDMA determines the effective SKRs.



3.Results and Discussion

Result 1: Baseline Performance with Fixed TDMA

Equal time allocation under dynamic cloud conditions

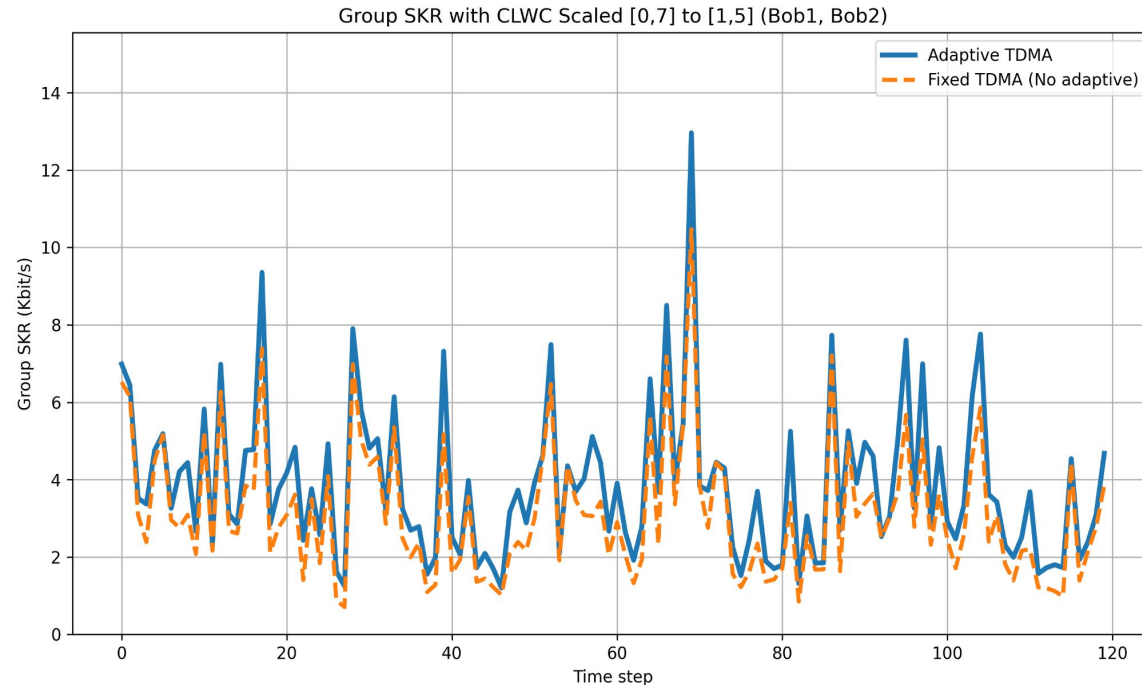


$t_1=t_2=0.5$

$$\text{Group SKR} = \frac{\min(R_1, R_2)}{2}$$

The worse Bob becomes the bottleneck under fixed equal allocation.

Result 2: Adaptive TDMA with CLWC Capping and Scaling

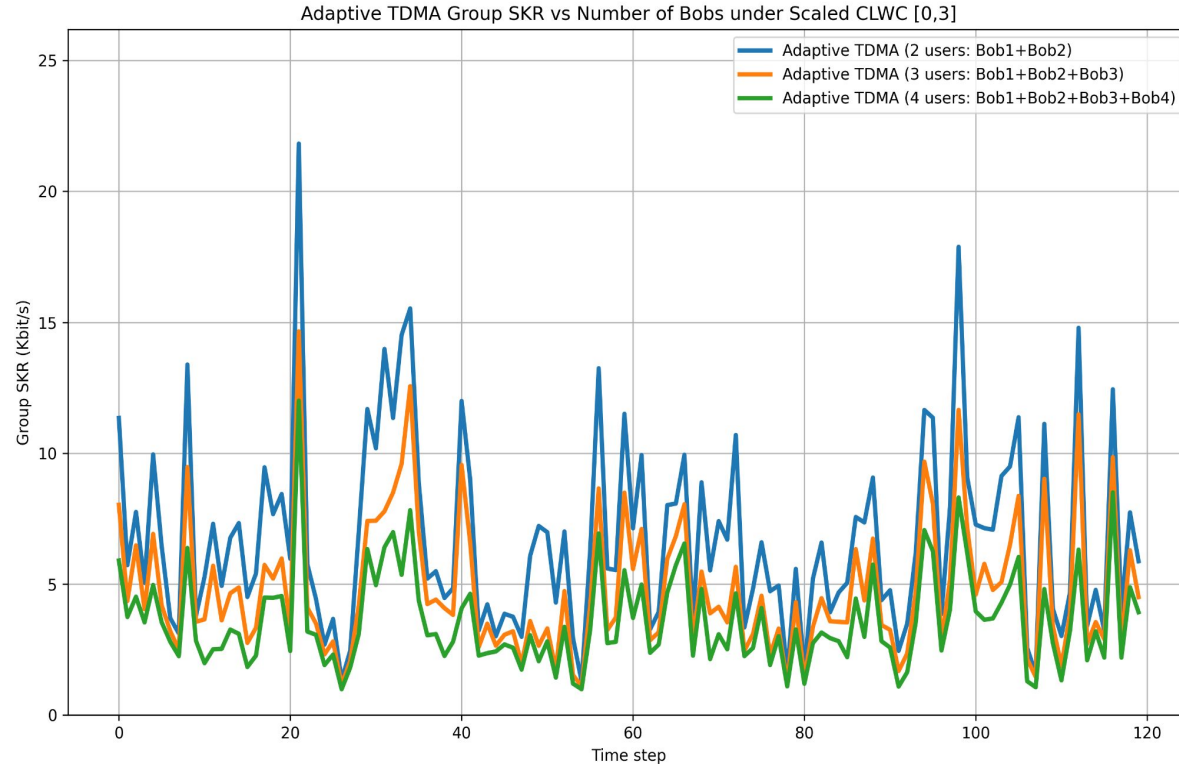


- CLWC is capped at 7 and scaled from $([0,7])$ to $([1,5])$ for TDMA scheduling.
- The lower bound prevents extremely small time allocation to a clearer Bob.
- Adaptive TDMA allocates more time to the worse channel and achieves a higher Group SKR than Fixed TDMA.

Key message:

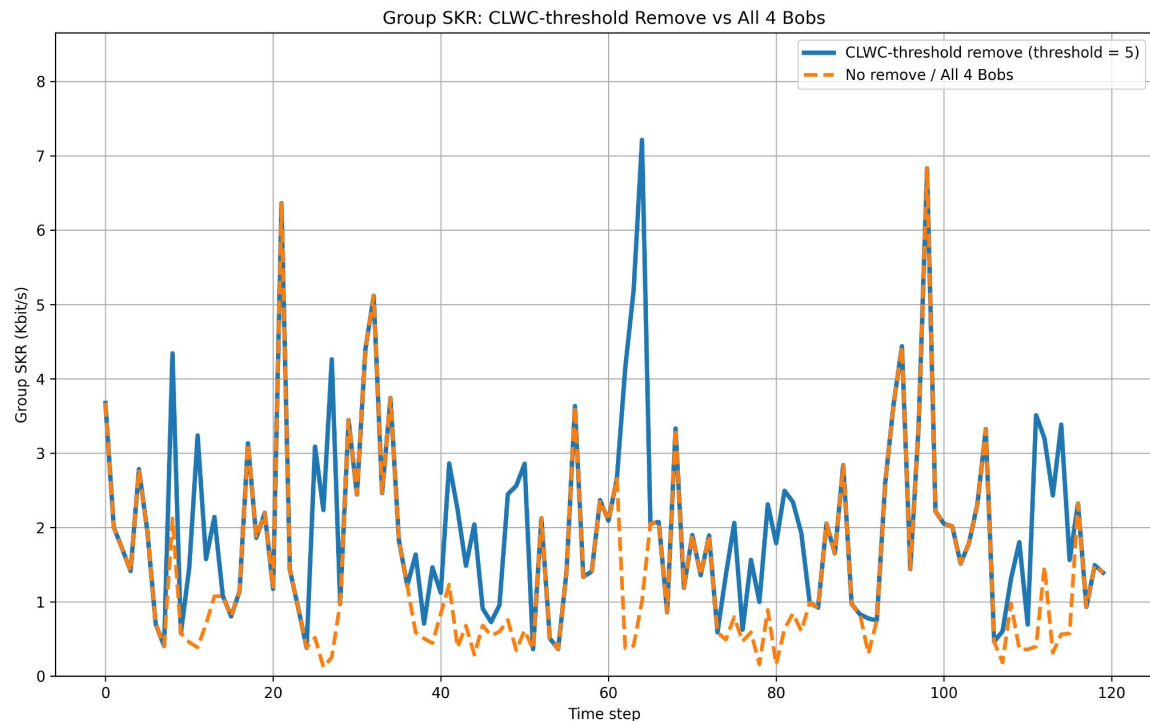
Adaptive TDMA improves Group SKR while avoiding excessively unbalanced time allocation.

Result 3: Effect of Group Size on Group SKR



As the group size increases, the Group SKR decreases because the worst Bob is more likely to limit the group.

Temporary Worst-Bob Exclusion Under Severe Cloud Conditions



Policy:

If any Bob has $CLWC \geq 5$,
temporarily exclude the lowest-SKR Bob.
TDMA slots are reallocated
among the remaining active Bobs.

Trade-off:

Can improve the active-group SKR,
but no new key for the excluded Bob.

The excluded Bob is re-evaluated at the
next time step.

4. Conclusion

Conclusion

Key Findings

- Moving clouds create different and time-varying SKRs for each Bob. The worst Bob limits the Group SKR.
- Adaptive TDMA with CLWC capping and scaling improves Group SKR compared with fixed equal allocation.
- As the group size increases, the Group SKR decreases because poor links are more likely to affect the group.
- Temporary worst-Bob exclusion can improve the active-group SKR under severe cloud conditions.

Adaptive scheduling can improve the robustness of HAP-based multi-user QKD under dynamic cloud conditions.